Technical Notes

Roadmaps and Flowcharts

Sample training course outlines

Report from Betty Wilson

Report from Esther Prellis

Control Chart constants

"Z" table

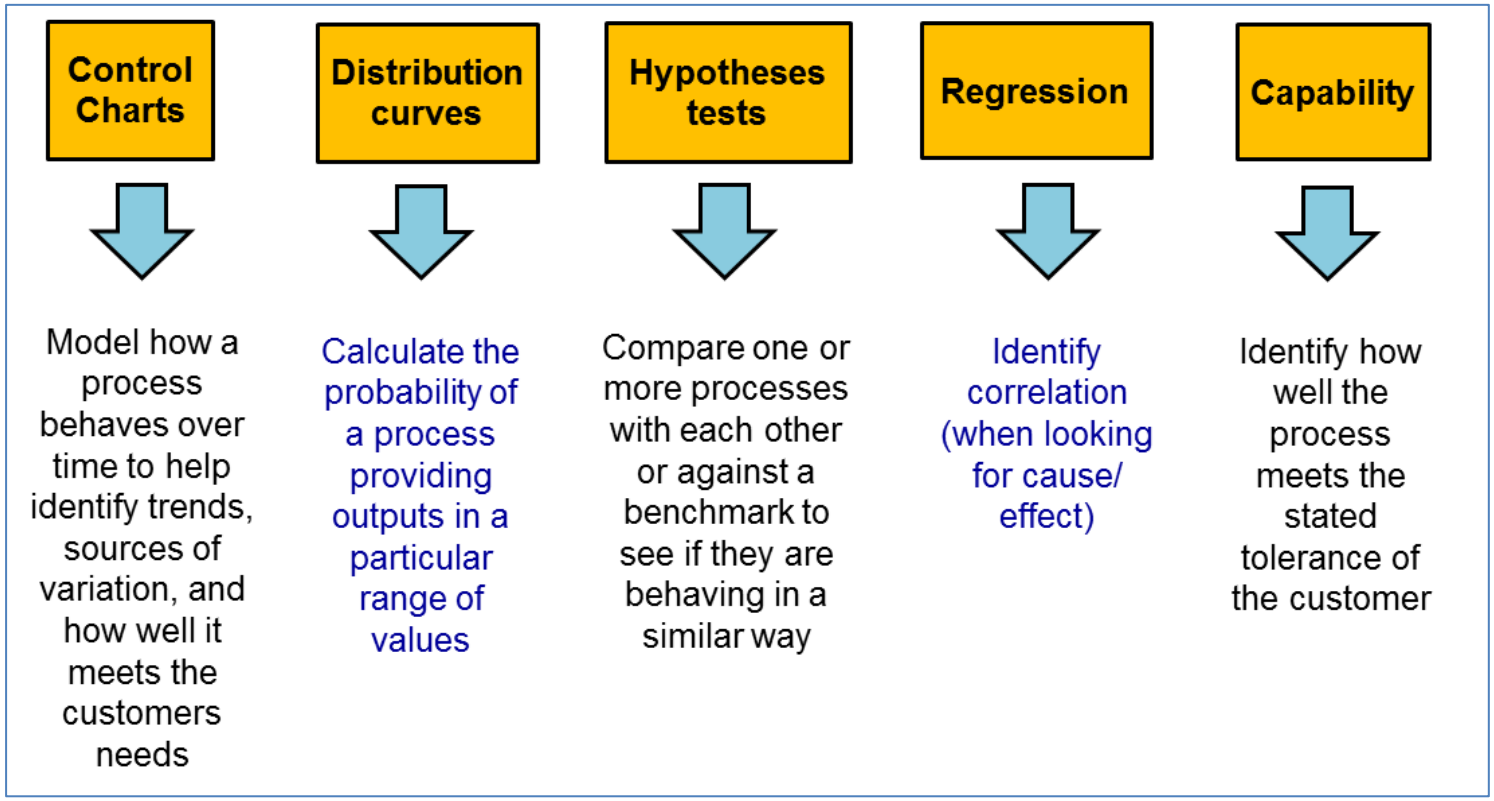
Stakeholder Management overview

Planning& estimating complex projects

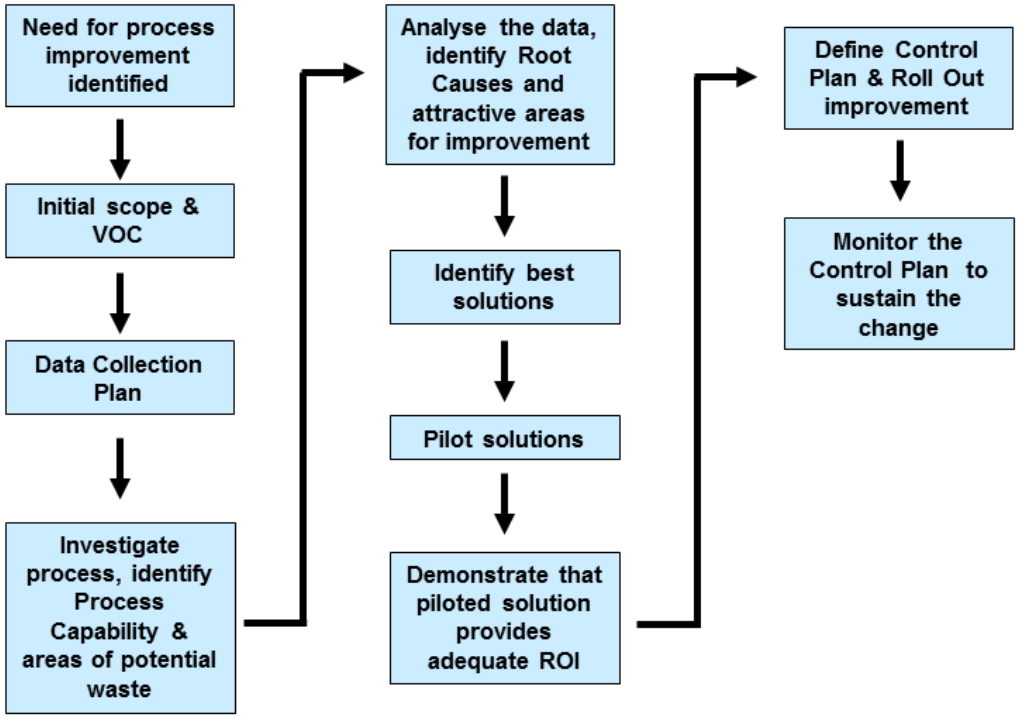
Business Case

Note: *with the exception of material taken from the slides this section is NOT considered as part of the exam BOK and you will not be tested on it.*

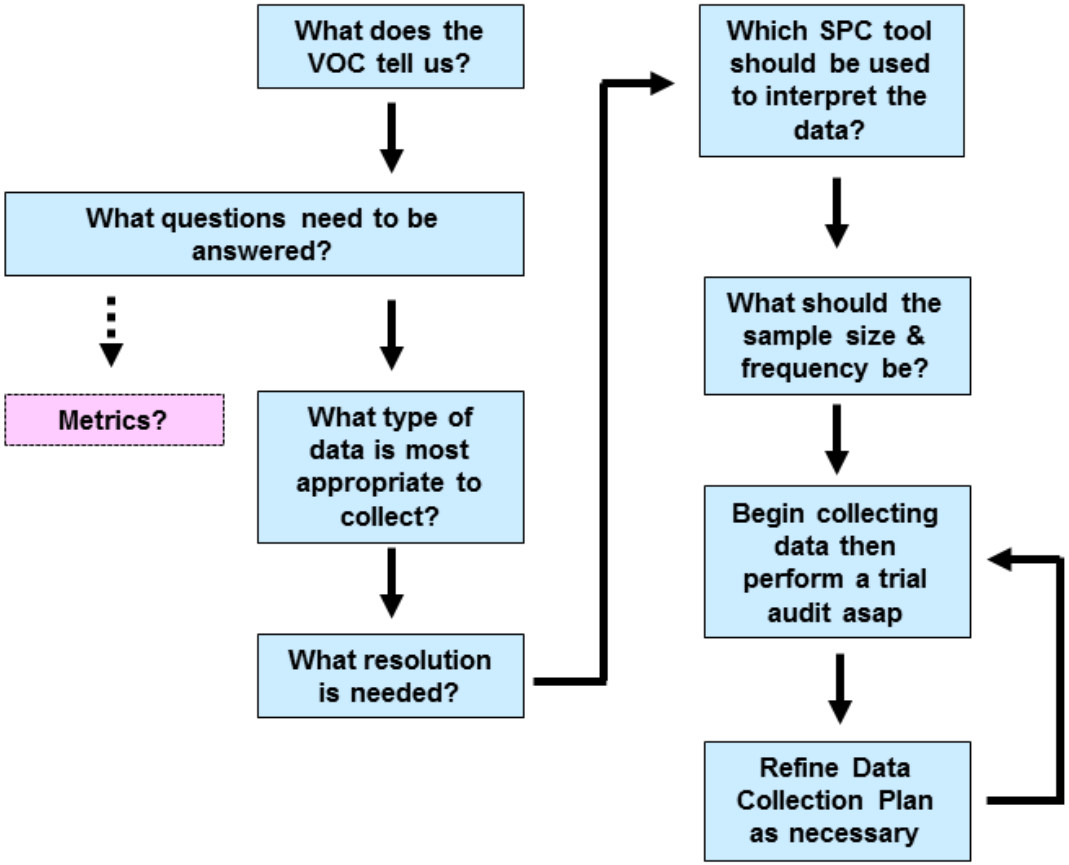
Top level statistical tool roadmap



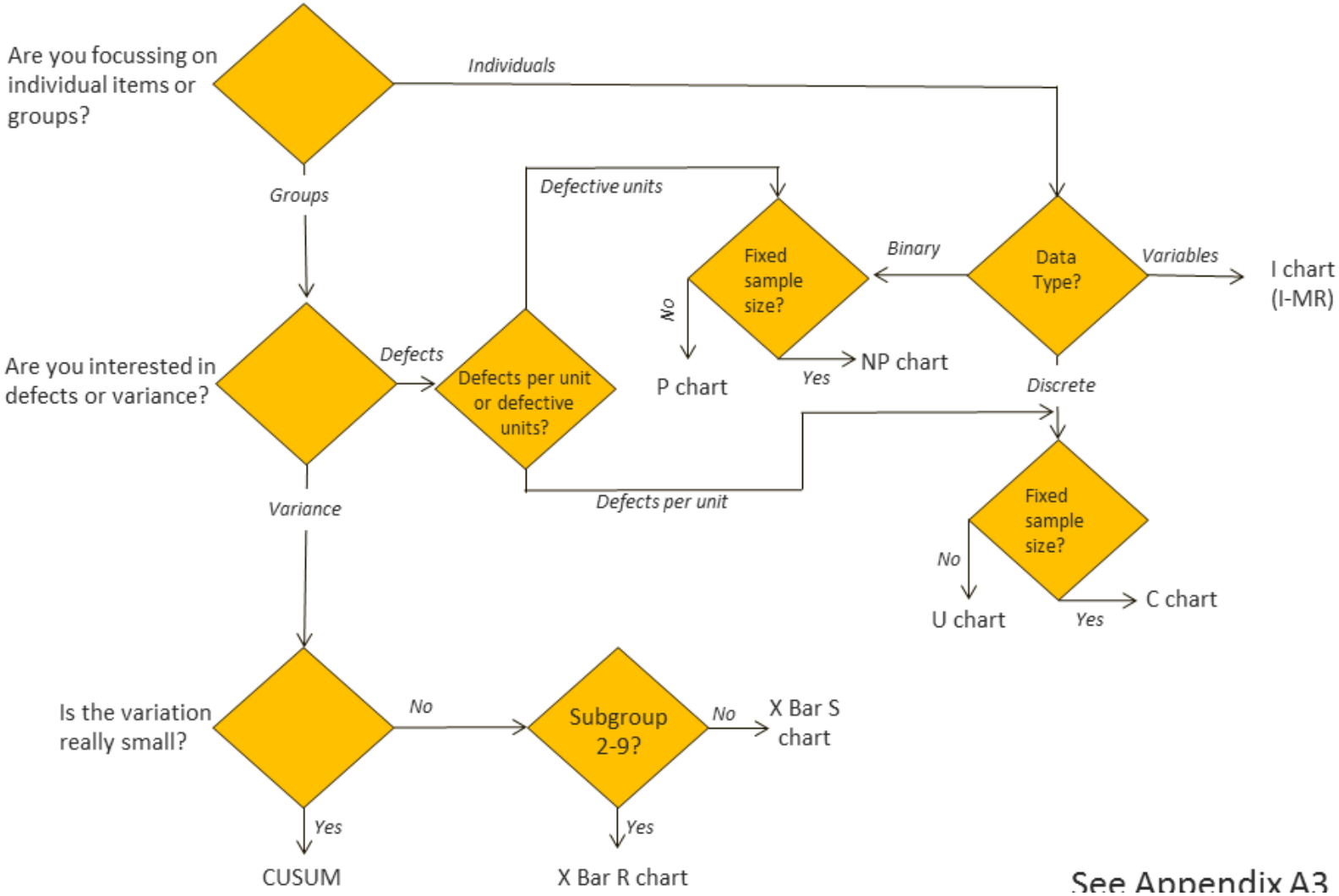
Basic Process Improvement methodology



Building the Data Collection Plan



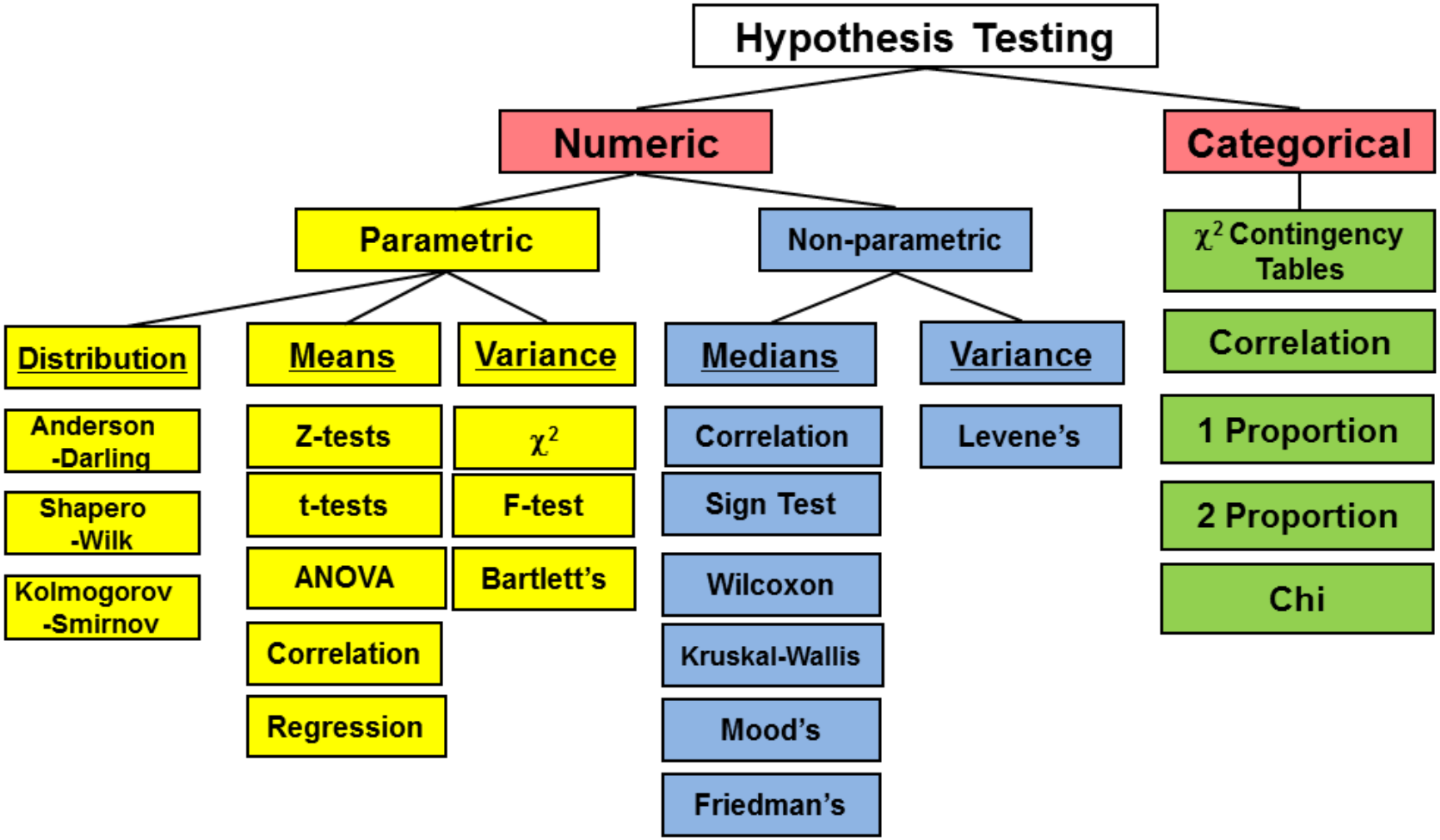
Control Chart selection roadmap



See Appendix A3

Hypotheses road maps

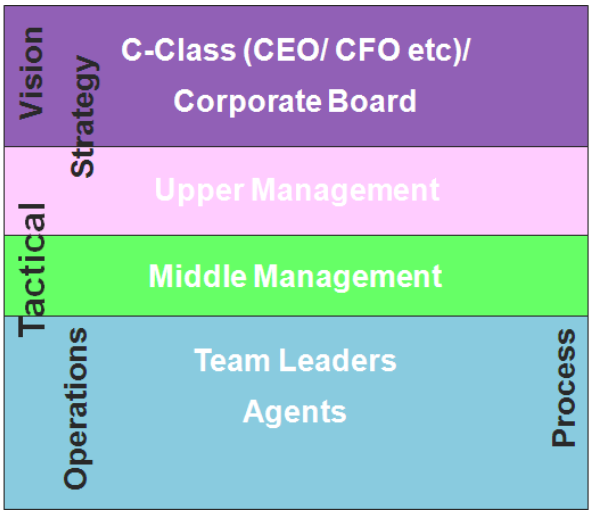
Comparing	Parametric test(s)	Non-parametric test(s)
The mean of a sample against a historical or known benchmark	One sample t-test F-test One proportion test	One sample sign test
The means of two (independent) samples	Two-sample t-test Two proportions test	Tukeys' test Mann-Whitney test Kruskal-Wallis test Mood's median test
Two related samples (before and after)	Paired t-test	Wilcoxen signed rank test
Means of 3+ (independent samples)	One-way ANOVA Bartlett's test	Kruskal-Wallis test Mood's median test Levene's test Chi-Squared test


Hypothesis Testing
Numeric
Categorical
Parametric
Non-parametric
χ^2 Contingency Tables
Distribution
Means
Variance
Medians
Variance
Correlation
Anderson -Darling
Shapero -Wilk
Kolmogorov -Smirnov
Z-tests
t-tests
ANOVA
Correlation
Regression
χ^2
F-test
Bartlett's
Correlation
Sign Test
Wilcoxon
Kruskal-Wallis
Mood's
Friedman's
Levene's
1 Proportion
2 Proportion
Chi

Sample Training Course outlines

These outlines provide a basic framework for Lean 6 Sigma training within an organisation.

Modules will need to be modified and may be combined as necessary.



Courses / Modules

Process White Belt
C-Class White Belt
Yellow Belt
Green Belt
Company Champions
Board Level Champions
Lean management – Middle Management
Lean management – Upper Management

Process White Belt

Time 2-4 hours

What is Lean 6 Sigma?
History
Philosophy
What it will mean to you
Why it is to be introduced
The Belt Structure
Company Belt structure (include role holders)
The Champion Framework
Company Champion Framework (as known)
Board Level Champion
Local Implementation schedule
How to get more information / volunteer
Lean game

C-Class White Belt

2hour presentation

Overview of Lean 6 Sigma
History and major practitioners
Philosophy
Changes to staff working practice
Benefits
Cost of poor quality
Cost of poor process sigma
What is expected of senior managements
Gemba
MBWA
Corporate Lean Metrics
Corporate Implementation schedule

Yellow Belt

1 Day

Review of "What is L6S"
Seven wastes of Lean
Exercise: Waste in our department
Lean triad
Exercise
Kaizen
Appropriate improvement methodology (as selected for company)
Lean Teams (if used) if used
Exercise

Green Belt

3 days + Exam (can be split e.g. 3 consecutive Thursdays)

GB and BB and Champion responsibilities
DMAIC
Small Project Management (based on standard company practice)
The Lean PM
Limits and Escalation
Process Flow and levelling techniques
Exercise: Process flow
Statistics for Green Belts
Exercise: Statistical Case Study

Company Champions

0.5 – 1 day

Aims of Lean 6 Sigma
Philosophy of Lean
Benefits of Lean 6 Sigma
Cost of poor quality
Cost of poor process sigma
Other benefits
GB and BB and Champion responsibilities
Importance of quick escalation
Selection and project oversight (small projects)
Boundary and escalation
Exercise: Project selection

Board Level Champions

1-2 days best as a contiguous block

Assumes C-Class White Belt Course

Importance of quick escalation
Project / Programme selection
Care and feeding of Black Belts
Developing and supporting the Lean 6 Sigma culture
Governance and risk
Corporate Lean metrics

Lean management – Middle Management

1-2 Days

Lean 6 Sigma corporate goals
Lean 6 Sigma culture
Your part in this
Seven wastes of Lean
Lean triad
Kaizen
Appropriate improvement methodology (as selected for company)
Overview of SPC
Corporate Lean Metrics
Division / Department metrics
Characteristics of Lean Managers
The Lean view of Middle Managers
Lean Cells

Lean management – Upper Management

1-2 days

Overview of Lean 6 Sigma
History and major practitioners
Philosophy
Benefits
Cost of poor quality
Cost of poor process sigma
Other benefits
What is expected of senior management
Gemba
MBWA
Lean 6 Sigma corporate goals
Lean 6 Sigma culture
Your part in this
Seven wastes of Lean
Lean triad
Kaizen
Appropriate improvement methodology (as selected for company)
Overview of SPC
Corporate Lean metrics
Characteristics of Lean Managers
Lean Cells (overview)
Setting appropriate staff goals

Report from Betty Wilson

From: Betty.wilson@CS&FGroup.co.uk

To: Ed.Mason@CS&FGroup.co.uk

To: Nayan.Mongia@CS&FGroup.co.uk

Subject: Bag Fill issues production line A

Hi all

As requested here is the preliminary report on the Bag Fill problem.

There appears to be a problem with mis-wrapping and there appears to be a daily pattern.

It seems that when the line starts, bags tend to be overfilled and there is quite a big difference between individual bags. As the line warms up, both the over fill and the variance decrease. This morning it took about 2 hours to start settling down.

We saw a similar pattern after the lunch stoppage although the effect was shorter lived - perhaps this had something to do with the fact that the line was stationary but not powered off.

In conclusion, it seems that both variation and bag fill are worse when the line is first started, but gradually both decrease as the line runs. One days' worth of data is far too soon to commit ourselves, however the fill station on Line A seems to be in a worse mechanical state than Line B. One of the mechanics on the team has pointed out that the drive belts are different on A and B as well.

Another thing that has come to light is that the whole production line shut down for 30 minutes at lunch time! Not only did we stop making sweets but when the Line re-started we found that it was up to its old tricks again.

So here is our short term plan:

(1) We are investigating the drive belts and will consider replacing them, perhaps as part of a full service - it will take a couple of days to get this sorted so I will keep you informed.

(2) We suggest forming a RAT to look at removing the half hour lunch time stoppage which seems to have no real justification.

(3) We will continue measuring so that we can identify other areas which contribute to the problem Line A is experiencing.

I will keep you up to date on the belts/ service issue.

Regards,

Betty

Report from Esther Prellis

Write: Delivery complaints

File Edit View Insert Format Options Tools Help

Send | Spelling | Attach | Security | Save

From: Esther.P@CS&F.Co.UK

Reply-To: Esther.P@CS&F.Co.UK

To: Allan.F@CS&F.Co.UK

To: Nayan.M@CS&F.Co.UK

Subject: Delivery complaints

Body Text | Variable Width

Hi Alan

I've looked into the on line deliveries as promised and there do seem to be a few problems

We back-tracked over 20 days and the shippers were constantly missing agreed delivery slots

With the possible exception on a single day, there seems to be no specific cause just that the delivery company do not have the capability to meet their agreed targets

I have the facts and figures from our investigation and have schedules a meeting with our account manager. I have requested that someone more senior also attends as I have told them that I need to come out of the meeting with an action plan. I don't know if you or Nyan wishes to come with me?

I also think we need to be thinking of a replacement as our delivery partner - perhaps we can discuss this at the next board meeting?

Regards

Esther

Control Chart Constants

Sample Size	For Averages	For Ranges		Standard Deviation
n	**A_2**	**D_3**	**D_4**	**d_2**
2	1.88	0	3.29	1.13
3	1.02	0	2.58	1.69
4	.73	0	2.28	2.06
5	.58	0	2.11	2.33
6	.48	0	2.00	2.53
7	.42	0	1.92	2.70
8	.37	.14	1.87	2.85
9	.34	.18	1.82	2.97
10	.31	.31	1.78	3.08

Z Table

Z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.0000	0.0040	0.0080	0.0120	0.0160	0.0199	0.0239	0.0279	0.0319	0.0359
0.1	0.0398	0.0438	0.0478	0.0517	0.0557	0.0596	0.0636	0.0675	0.0714	0.0753
0.2	0.0793	0.0832	0.0871	0.0910	0.0948	0.0987	0.1026	0.1064	0.1103	0.1141
0.3	0.1179	0.1217	0.1255	0.1293	0.1331	0.1368	0.1406	0.1443	0.1480	0.1517
0.4	0.1554	0.1591	0.1628	0.1664	0.1700	0.1736	0.1772	0.1808	0.1844	0.1879
0.5	0.1915	0.1950	0.1985	0.2019	0.2054	0.2088	0.2123	0.2157	0.2190	0.2224
0.6	0.2257	0.2291	0.2324	0.2357	0.2389	0.2422	0.2454	0.2486	0.2517	0.2549
0.7	0.2580	0.2611	0.2642	0.2673	0.2704	0.2734	0.2764	0.2794	0.2823	0.2852
0.8	0.2881	0.2910	0.2939	0.2967	0.2995	0.3023	0.3051	0.3078	0.3106	0.3133
0.9	0.3159	0.3186	0.3212	0.3238	0.3264	0.3289	0.3315	0.3340	0.3365	0.3389
1.0	**0.3413**	**0.3438**	**0.3461**	**0.3485**	**0.3508**	**0.3531**	**0.3554**	**0.3577**	**0.3599**	**0.3621**
1.1	0.3643	0.3665	0.3686	0.3708	0.3729	0.3749	0.3770	0.3790	0.3810	0.3830
1.2	0.3849	0.3869	0.3888	0.3907	0.3925	0.3944	0.3962	0.3980	0.3997	0.4015
1.3	0.4032	0.4049	0.4066	0.4082	0.4099	0.4115	0.4131	0.4147	0.4162	0.4177
1.4	0.4192	0.4207	0.4222	0.4236	0.4251	0.4265	0.4279	0.4292	0.4306	0.4319
1.5	0.4332	0.4345	0.4357	0.4370	0.4382	0.4394	0.4406	0.4418	0.4429	0.4441
1.6	0.4452	0.4463	0.4474	0.4484	0.4495	0.4505	0.4515	0.4525	0.4535	0.4545
1.7	0.4554	0.4564	0.4573	0.4582	0.4591	0.4599	0.4608	0.4616	0.4625	0.4633
1.8	0.4641	0.4649	0.4656	0.4664	0.4671	0.4678	0.4686	0.4693	0.4699	0.4706
1.9	0.4713	0.4719	0.4726	0.4732	0.4738	0.4744	0.4750	0.4756	0.4761	0.4767
2.0	**0.4772**	**0.4778**	**0.4783**	**0.4788**	**0.4793**	**0.4798**	**0.4803**	**0.4808**	**0.4812**	**0.4817**
2.1	0.4821	0.4826	0.4830	0.4834	0.4838	0.4842	0.4846	0.4850	0.4854	0.4857
2.2	0.4861	0.4864	0.4868	0.4871	0.4875	0.4878	0.4881	0.4884	0.4887	0.4890
2.3	0.4893	0.4896	0.4898	0.4901	0.4904	0.4906	0.4909	0.4911	0.4913	0.4916
2.4	0.4918	0.4920	0.4922	0.4925	0.4927	0.4929	0.4931	0.4932	0.4934	0.4936
2.5	0.4938	0.4940	0.4941	0.4943	0.4945	0.4946	0.4948	0.4949	0.4951	0.4952
2.6	0.4953	0.4955	0.4956	0.4957	0.4959	0.4960	0.4961	0.4962	0.4963	0.4964
2.7	0.4965	0.4966	0.4967	0.4968	0.4969	0.4970	0.4971	0.4972	0.4973	0.4974
2.8	0.4974	0.4975	0.4976	0.4977	0.4977	0.4978	0.4979	0.4979	0.4980	0.4981
2.9	0.4981	0.4982	0.4982	0.4983	0.4984	0.4984	0.4985	0.4985	0.4986	0.4986
3.0	**0.4987**	**0.4987**	**0.4987**	**0.4988**	**0.4988**	**0.4989**	**0.4989**	**0.4989**	**0.4990**	**0.4990**

The value of "Z" is selected by using a combination of the top row and the left hand column of the table. This allows you to specify “Z” to **two** decimal places.

Notice values in the LH column identify Z to **one** decimal place. The values on the top row give the **second** decimal value.

The values in the main table are probabilities that for a population proportion between the mean and the value specified by Z.

Example:

The proportion of the population between the mean and 1 SD (Z=1.00) = approx. 34%

The proportion of the population between the mean and 1.04 SD (Z=1.04) = approx. 35%

Stakeholder Analysis

Basic terminology

Stakeholders	Person or role with power over and/or interest in work to be done or its deliverables A person who has one or more concerns about the system to be built (because they own it, manage it, use it or other reason).
Concerns	Something that a stakeholder needs to know about a system A general kind of requirement (e.g. availability, usability) that is important to one or more stakeholders in the system, and determines the acceptability of the system to those stakeholders. A concern may be addressed in several view points
View point	A collection of concerns
View	The information provided by the EA to satisfy the questions of a View point

Definition of Stakeholders

There are many definitions of what a stakeholder here is one:

"Anyone who has an interest in, or may be affected by, the issue under consideration."

(British Computer Society)

So in summary Stakeholder(s) are:

- Interested in a project
- Affected by a project
- Can influence a project

What is Stakeholder Management used for?

Stakeholder management is usually identified in the initiation and includes the following:

- To determine levels of power and Interest of key stakeholders
- Plan fact finding and communication strategy
- Understand – analyse needs from fact finding
- Manage expectations

Why we need Stakeholder management

Stakeholder management allows architects to win the support of stakeholders thus enabling us to define architecture successfully due to the following benefits:

- Powerful stakeholders can be identified upfront which means that we can uses their input to shape the architecture which enables architects to improve the quality of the architectural model procedures
- Support from powerful stakeholders cab help to win more resources
- Knowing your stakeholders and communicating with them regularly can lead to a more clear and concise detailed architecture
- Increased anticipation of reactions to reactions to the architecture means that architects are more likely to anticipate changes as react in positive manner
- Architects can identify competing and conflicting objectives early on and develop a strategy to deal with them early on

A stakeholder methodology

The following is a 4 step way to handle Stakeholders advocated by TOGAF:

1. Identify Stakeholders
2. Classify Stakeholder Positions
3. Determine Stakeholder Management Approach
4. Write up the Communication Plan

(1) Identifying stakeholders

It is key that Stakeholder are identified upfront by architects. It is often recommended to brainstorm as many relevant stakeholders up front. When brainstorming the following questions should drive stakeholder identification:

- Who is at benefit and disadvantage from this change?
- Who controls the change?
- Who makes the decisions?
- Who controls the resources?
- Who has the specialist skills the project needs?
- Who has influence?

Categorising Stakeholders

There are various categories of stakeholders. A list of generic Stakeholder categories is listed below

- Owners
- Customers
- Suppliers
- Employees
- Managers
- Partners
- Regulators and legal bodies
- Competitors

There are three key types of Stakeholder:

- Project Stakeholder
- Business Stakeholders
- External Stakeholders

Project Stakeholders

- Project Manager- Day to day control
- Business Analysts - Elicitation & Analysis
- Solution Developers - Technical feasibility, Produce prototypes
- Testers - validating that requirements are testable
- Architects

Business Stakeholders

- Project Sponsor - Ultimately paying for project, Signs off the requirements
- Subject matter expert - Local SME :Understands the business, Domain SME: Understands the industry
- End users and managers: For whom the solution is designed

External stakeholders

- Customers
- Regulators
- Suppliers - Products and services
- Competitors - Can be difficult to engage with

(2) Classifying Stakeholders Positions (Stakeholder Analysis Matrix)

Below is an example of classifying stakeholders based on their commitment and understanding. There are three key rankings high, medium and low.

Stakeholder Group	Stakeholder	Ability to disrupt change	Current Understanding	Required Understanding	Current Commitment	Required commitment	Require support
Line Manager	Sara Jane	M	M	M	H	M	M
CEO	Mark Johnson	H	M	H	H	M	M
CFO	Christopher Smith	H	L	M	L	M	H

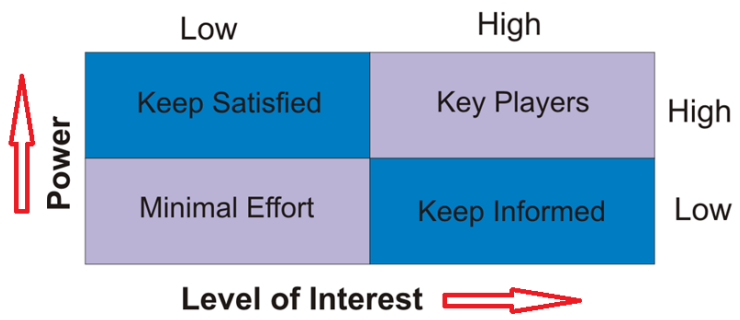
(3) Determine Stakeholder Management Approach (Power/Interest Matrix)

When identifying and classifying stakeholder we can end up with a long list of stakeholders of which some can have the power projects in either a positive or negative manner. So working out power, interest and influence can help focus stakeholder engagement to get key information that is vital to the project. The following technique will enable architects to devise the correct strategy for maintaining and engaging in stakeholder relationships.

First identify your stakeholders by the **Power** (influence they have over the project) and **Interest** (what their information needs are) using the following table as a guide:

	High	**Some**	**None**
Power (to influence the project)	Can change the direction of the project Tend to be directors or senior management level of affected area Make project decisions	Has some influence Would be listened to (possess specialist knowledge) Lower level managers, senior managers of non-affected areas or supervisory level Have the ear of those making decisions	Have (or feel they have) no ability to influence the project End users typical
Interest	Directly affected or impacted by the project Is of a major concern Employees in affected area Users	Indirectly affected or impacted by the project Not sole concern, more pressing or many other concerns Employees in a related area	Not affected by the project

Next construct a **Power/Interest grid** and plot your stakeholders onto it.



As a guide to the categories I have included the following table:

P/I Category	Information needs
Keep satisfied	These are Senior Managers & Leaders (think Board Members) who need to be kept abreast of matters but from a high level
Key Players	This group are likely to be more involved with the project possibly on a day-day (or at least frequent) basis. The Project Sponsor would fit into this category
Keep Informed	Someone who will be involved in the work but does not have the authority to stop or change the work. Someone working in the programme or project for example.
Minimal Effort	People who cannot stop the work and have very few information needs.

Roles are *cumulative* so, for example, if Sue is a member of the Board ("Keep Satisfied") and because of another role she performs is also categorised as "Keep Informed" she goes in the "Key Player" box in the power interest matrix.

(4) Write up the Communications Plan

The Communications Plan identifies Stakeholders and their communications needs:

- What they want to know
- When they need to know it
- How they need to be informed (Meetings, E-mail etc)

Typical things to include in your plan are:

- Escalations
- Gateways
- Team meetings
- Compliance & external reviews

Planning and Scheduling Complex Projects

Complex projects and programmes need more by way of planning than just simply tacking a list of milestones to a wall. There are three useful planning/scheduling tools that can be of use:

- Gantt Chart
- Critical Path Analysis
- PERT Charts

These tools provide the basis both for preparation of a schedule, and of resource planning (in small projects you might not all or any of them) .

During management of a project, they allow you to monitor achievement of project goals. They help you to see where remedial action needs to be taken to get a project back on course.

Gantt Charts tend to be used within projects while CPA and PERT charts tend to be used for more complex work and programmes, however remember this is just a rule of thumb – if you need them then use them.

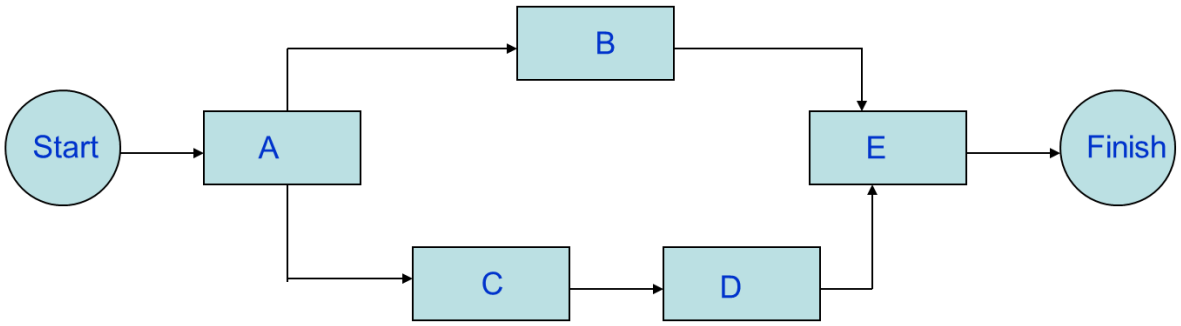
Networks

In project planning terminology, the term "**Network Diagram**" has a different meaning to IT. The project Network Diagram shows activities which need to be conducted serially or sequentially and those that can run in parallel and the events associated with them. A project defines an activity as a task that must be performed if the project is to be successful and an event as a milestone indicating either the start or completion of one, or several activities.

There are two main philosophies when constructing network diagrams, **Activity-on-node**, and **Activity-on-Arrow**

Activity-on-node

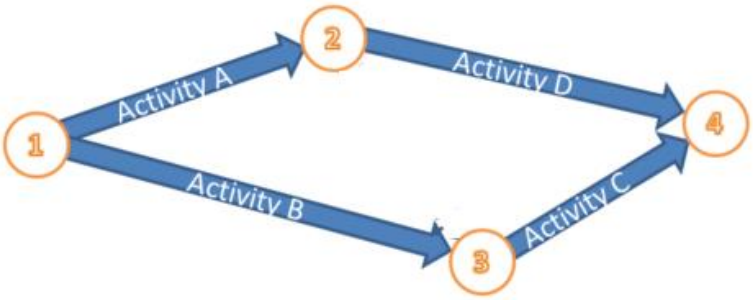
Activity-on-node is a project management term that refers to a precedence diagramming method which uses boxes to denote schedule activities. These various boxes or "nodes" are connected from beginning to end with arrows to depict a logical progression of the dependencies between the schedule activities. Each node is coded with a letter or number that correlates to an activity on the project schedule.



Typically, an activity-on-node diagram will be designed to show which activities must be completed in order for other activities to commence. This is referred to as "finish-to-start" precedence – meaning one activity must be finished before the next one can start. In the diagram above, activities A and D must be done so that activity E can begin. It is also possible to create other variations of this type of diagram. For example, a "start-to-start" diagram is one in which a predecessor activity must simply be started rather than fully completed in order for the successor activity to be initiated.

Activity-on-arrow (Arrow diagramming method or ADM)

In ADM the activities are shown as arrows



With the ADM network drawing technique, the start and end of each node or event is connected to an arrow.

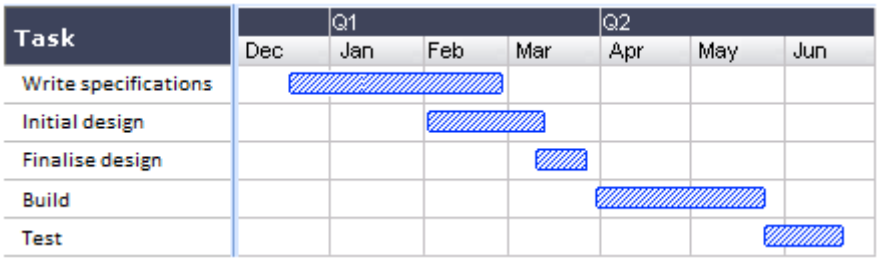
The start of the arrow comes out of a node while the tip of the arrow goes into a node. Between the two nodes lies an arrow that represents the activity.

The event represented by the circular node consumes neither time nor resources.

- A node is a specific, definable achievement in the project.
- It has zero duration and consumes nil resources.
- All activities that lead into a node must be completed before the activity lines following this node can start.

Gantt Charts

A Gantt chart is a well-known and easy to understand planning tool. It graphically illustrates the relationship between tasks/ activates and time:



As you can see above, the Gantt chart lists the tasks/activities on the left then shows when these will take place.

Gantt charts illustrate:

- Activity start points, end points, duration
- Whole project start points, end points, duration
- Overlap (resource contention?)

A Gantt chart graphically illustrates both the activities to be performed and how these are scheduled (you do not need much more to make a roadmap).

Critical Path Analysis (CPA)

Critical path analysis is a technique to construct a model of the project that includes the following:

- A list of all activities required to complete the project (also called a *work breakdown structure*)
- The duration of each activity
- The dependencies between the activities

What is the critical path?

The critical path could be defined as the least amount of time necessary to complete a number of tasks. Critical Path Analysis formally identifies tasks which must be completed on time for the whole work stream (in our case a project or programme) to be completed on time.

CPA allows identification of the following:

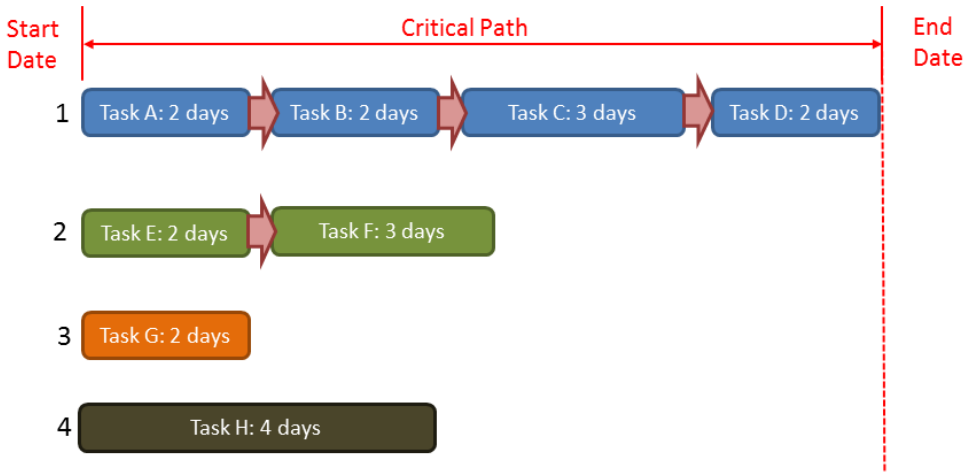
- The minimum time needed to perform a work stream
- Which activities or tasks could be accelerated to bring the end date forward
- Which activities and tasks can be delayed without endangering the end date
- Which activities and tasks can be run in parallel
- Whether you have a robust and realistic plan.

The downside of CPA is that it is not as intuitive as a Gannt chart partly as the relationship of tasks to a timeline are not really indicated.

Sequential and parallel tasks

A basic concept of CPA is that some activities are *dependent* on other tasks completing first (the concept of dependency has been mentioned before). Sequential tasks are tasks which need to be completed in order with each one finishing (more or less) before the next starts. Other activities have no dependencies and can be started at any time (so long as they have time to complete before the end date) – these are known as **parallel** tasks (in this section I will use "*tasks*" and "*activities*" interchangeably).

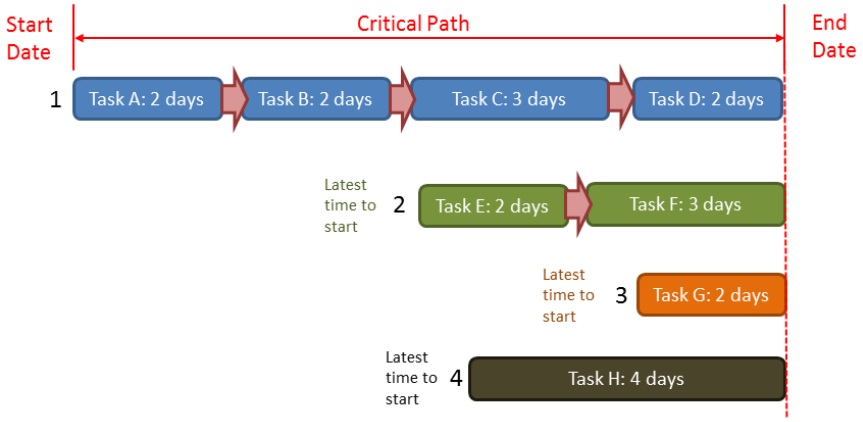
So a work stream such as a project or programme could consist of a mixture of parallel tasks and sequential chains. The length of the longest serial chain represents the *critical path (* the least amount of time you need to complete a work stream). Delaying any task on the Critical Path will delay the end date of the whole work.



Consider the diagram above – imagine that this illustrated a project which had two sequential chains (1 & 2) and two parallel tasks (3 & 4). As all of these four are not dependent on each other they can be started at any convenient time. 2-4 all take less time than 1 so chain 1 sets the minimum time this

project will take to complete. If Task B extends to 3 days then the end date moves back a day accordingly. If Task E extended by 2 days then the end date would remain unchanged. Chain 1 defines the critical path for this project, as tasks 2-4 are not dependent on each other or on 1 they can "float" that is they can start after 1 and can expand without harming the end date. However each has a date that they must be started on otherwise the project will overrun.

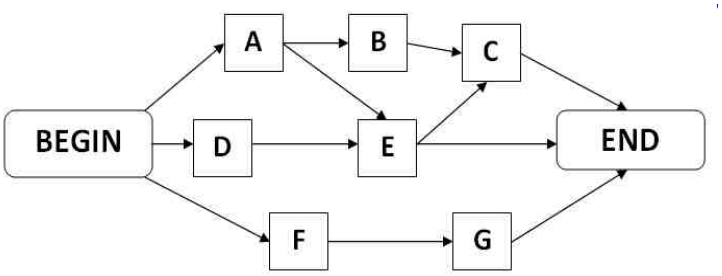
Here is the same project illustrating this state of affairs



I suspect at this point you can begin to see the value that CPA might add to your planning activities!

Activity networks

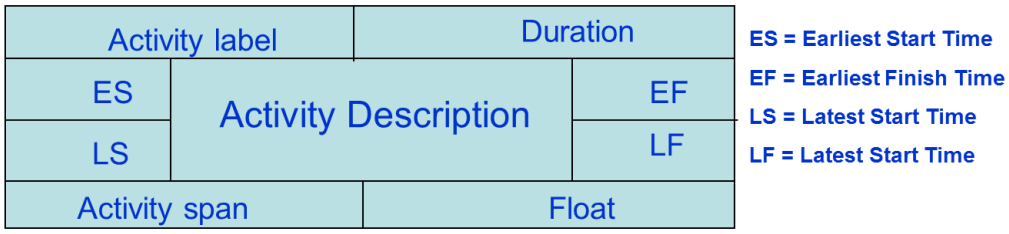
CPA uses **Activity Networks** as a way of visually linking together the activities needed to produce a result. Each node has some useful planning information attached to it (as I will describe a little later).



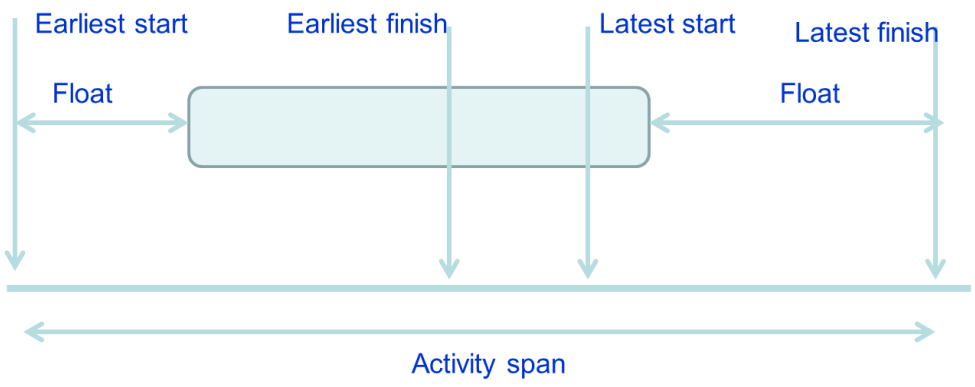
Each box (**node**) on the network identifies an activity, the arrows (**connectors**) show the dependency and flow. The network should start with "Begin" and "End" boxes

Activity boxes

It is common to add extra information into each activity box to aid planning:



You can see how useful some of these are from the following diagram:



Program Evaluation and Review Technique (PERT)

PERT is a method to analyse the tasks involved in completing a given project, especially the time needed to complete each task, and identifying the minimum time needed to complete the total project.

PERT originally was an activity on arc network, in which the activities are represented on the lines and milestones on the nodes. Over time, some people began to use PERT as an activity on node network. For this discussion, we will use the original form of activity on arc.

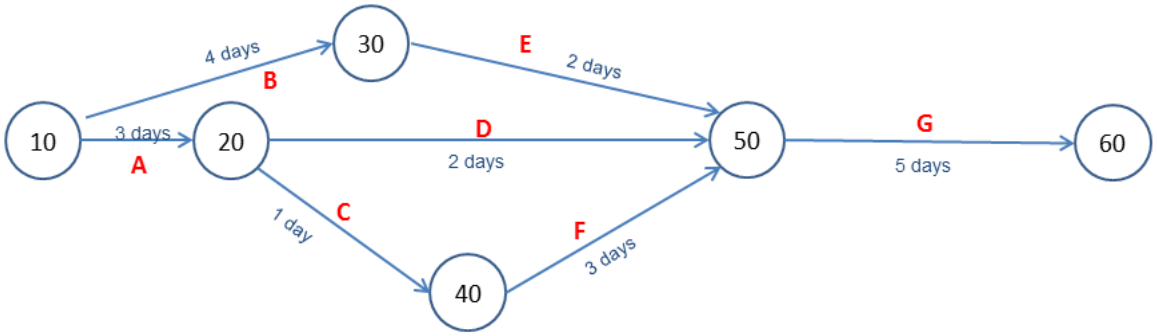
The Program Evaluation and Review Technique (PERT) is a network model which was developed in the late 1950s for big complex projects (the Polaris Missile project actually). PERT represents activities by arcs (lines with arrows) and milestones by nodes (Circles).

While PERT charts are usually complex we need to just consider a very simple example:

> TomSoft an internet gaming company is developing a proof of concept new game “Car Crash 3000” as part of an AGILE project.
>
> The game involves players logging on and creating a character, they then select a car and the appropriate bits (these are *actors*) then drive round a rendered 3D map committing mindless acts of mayhem and having a thoroughly good time (OK you’ve got the picture!).
>
> The first SPRINT has the seven activities as listed below:

Activity	**Estimated time**	**Depends on task ..**
A: Create basic actors	3 days	-
B: Write the wire frame map	4 days	-
C: Build log in screen	1 day	A
D: Write the player inventory database	2 days	A
E: Write the rendering (3D code)	2 days	B
F: Build player inventory system	3 days	C
G: System & User testing	5 days	D E F

And a PERT chart illustrating this on a network diagram would look like this:



The nodes are numbered, the higher the number the later the activity needs to start. It is common to leave gaps in the numbering so that new activities can be easily inserted (the numbers are ordinal not relative).

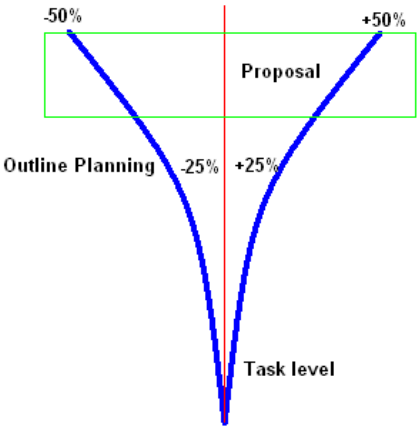
The arrows indicate dependencies, tasks, and the expected duration of each task.

Estimating times

"An Estimate is a probabilistic assessment based on Skill and Experience of the Time and Resources required for the Delivery of a Specified End Product."

Estimating is both an art and a skill, unfortunately we do not have time to go into estimation methods here a simple listing of common techniques will have to satisfy us:

Cone of Uncertainty



At the beginning of a project little is known about the work, and therefore estimates are subject to high variability

Beginning:

100% estimate range

Outline Planning:

50% range

Task: Single estimate

Compare estimations based on work

Bottom-up & Top-down estimating

Bottom-up

Usually performed at the task level

- Identify every discrete activity in a stage of the Project
- Estimate for each activity in the stage
- Sum activity estimates into a total estimate for the stage

Top-down

Usually performed when doing high level or overall planning

- Estimate for the Project overall
- Typically only a single figure produced
- Identify the major stages required
- Check the relative sizes of stages
- Apportion the overall estimate
- Usually based on our knowledge of past cases

Expert Judgment (wide-band Delphi)

- Gather experts
- Pose the question (e.g. “How long will it take”)
- Obtain estimates from each expert individually
- Let estimates be seen and discussed by the group
- Get justification for those outside the norm
- Allow people to change their minds
- Arrive at consensus

A good example of wide-band Delphi is **Planning Poker**

Analogy

Estimate produced by comparing the present project with previous completed and successful projects.

Advantage:

Based on actual experience

Disadvantage:

Is the experience valid?

Is the experience recorded?

One useful thing about PERT is that it can handle uncertainty in estimations. For each activity it is possible to indicate any of three types of estimation:

Optimistic time	Usually the shortest time we estimate the activity to take
Most likely time	The completion time having the highest probability **Note:** Do not confuse *Most likely time* with *Expected Time*
Pessimistic time	The longest time we reasonable can foresee the activity taking

Expected Time

PERT uses a statistical formula (the *beta probability distribution)* to estimate Optimistic, Most likely, and Pessimistic, times. We do not need to go into this here however the **Expected Time** (that is the time we reasonable can assume the task will take) can be approximated using the following formula:

Expected Time = [Optimistic + 4(Most likely) + Pessimistic] / 6

This equation is a weighted average where the most likely estimate is weighted 4 times more heavily than the optimistic and pessimistic estimates. This prevents the PERT number from being too heavily skewed in one direction.

Example:

Suppose a project manager wishes to estimate time to complete some work. She sits with the team and using their experience they estimate the following:

1. The time it would take to perform the work if there were no hold ups at all (let's say this comes to 7 days)
2. Common and frequently occurring hold-ups in work of this nature and how much they could add to the project duration (5 extra days)
3. Unlikely and rare hold-ups that could occur in work of this nature and how much these would add to the project duration (17 extra days)

You can see that experience and estimation techniques are critical here!

From this the team can calculate:

- The most likely time to complete the project (1+2 from the above list). This will be 12 days in our example.
- The most optimistic time for the project (1 only) if there are no delays (7 days).
- The most pessimistic prediction for the project (1+2+3). This is 29 days.

However, while we are pretty sure that some of the hold-ups will happen, we are confident that not all of them will. So what would a reasonable estimate of project duration be?

Using the formula described earlier, The PERT estimate for the duration of the project would be:

[7 + 4(12) + 29]/6 = 84/6 = 14 days

To calculate the variance for each activity completion time you can use the following approximation

(Expected - Optimistic) / 6

Summary Steps in the PERT Planning Process

1. Identify the specific activities and milestones.
2. Determine the proper sequence of the activities.
3. Estimate the time required for each activity.
4. Construct the network diagram.
5. Determine the *critical path*. (discussed earlier)
6. Update the PERT chart as the project progresses to reflect changes.

What PERT will illustrate

PERT can help identify the following:

- ☐ Expected project completion time.
- ☐ Probability of completion before a specified date.
- ☐ Start date to complete on a given date
- ☐ Sequential & parallel paths
- ☐ The critical path activities that directly impact the completion time.
- ☐ The activities that have slack time and that can lend resources to critical path activities
- ☐ Activity start and end dates.

PERTS's limitations

Like all techniques PERT has areas that it does not perform well in. A major area of concern is the estimation system used.

Despite the statistical formula used estimation is always going to be just that – a hostage to fortune so there is still a need to have experienced estimators. Group estimates (at least 3 people help) but there is always bias however hard you try to avoid it.

Even if the estimates are sound the distribution may not be exactly as assumed by PERT so there might be in-exactitude here.

PERT focuses on one chain as being the critical path but cannot easily compensate for other chains becoming critical as they experience delays and because of this PERT often underestimate actual project completion time (**Monte Carlo Simulations** technique can be used to compensate for this problem in some degree).

The Business Case

Any change to a process causes disruption, inconvenience, and takes time and money. Why should we want to put ourselves through this aggravation unless it was justified – that is the changed state produces a beneficial outcome which outweighs the trouble caused? Consider the hassle of going on holiday. All the effort: the packing, finding the passports, the ghastly queues at the airport full of screaming kids (fortunately I'm passed that stage now – mine slope off to the bar!), not to mention the cost! Why do you do it? Because despite all the trouble you enjoy your break, you look back at it with fond memories; simply put - you get out of your holiday more than you put in.

In business every proposed change should be justified to demonstrate that it is worthwhile. The formal way of justification is to create a **Business Case**. Business Cases are many things to many people, at a minimum you will probably wish to include the following:

- ROI (Return On Investment)
- Options (business or technical),
- Impacts
- Risks

A Business Case provides justification for a change to take place; this means it justifies the work to make the things (*deliverables*) that cause the change (a new process for example) and the work to plan the delivery and production of those deliverables. In other words a Business Case also justifies a project or a programme.

ROI (Return on Investment)

ROI is a bit of a balance sheet – on one hand you have the expected benefits of a change on the other you have the cost of making the change **plus** the ongoing maintenance and support costs. If the former outweigh the latter then the project is viable. If not the project should not go ahead. Benefits should be measurable in some way, cash (often savings made) is a good comparator as it can be easily stacked up against costs however this is not always possible so a benefit might be something like: "regulations complied with", or, "the resolution of specific problems" (such as the benefit of running a project to provide data integrity is to save the cost of data dis-integrity. In many cases a little forethought will allow you to take a non-cash benefit and convert it into a cash benefit for comparison reasons.

Cost-benefit analysis

A financial technique used to contrast the costs and benefits of a change. Often there are several ways to make a change and a Cost/Benefit analysis will help indicate which might be the most beneficial to an business.

One way of performing a Cost-benefit analysis is Discounted Cash Forecasting which converts a future return into current savings.

Options

Options in the Business Case should list alternative solutions to meet the need for change. A Business needs to increase its profitability – options could be: Increase Market Share, sell off a loss making subsidiary, or invest in re-engineering a process to make it more efficient. One option you should consider putting in a Business Case is the option to "Do Nothing", this acts as a nice illustration of what will happen if nothing is done!

How a selected Business or Technical option is achieved is described by **Solution Options** (discussed a little later)

Impacts

Describes the effect the change would have on the business should it be carried out. This should describe the work to be done and the changes made to the affected organisations

Solution Options

Once the Technical or Business solution has been decided it is necessary to identify how that solution can be achieved. For example, supposing that (from the previous example) it has been decided to sell off a loss making subsidiary, how would the business go about this? Should they call in a firm of consultants, find their own buyer and arrange a sale, or form it as a separate entity and float it? These are known as **Solution Options** (as opposed *to Business or Technical Options* in the Business Case)

At the **solution vision** stage there may be several Solution Options proposed. As the cycle progresses and more information becomes available, the options will be contrasted & compared. Comparison methods commonly employed could be:

- Cost-benefit analysis (covered previously)
- Risk analysis
- Gap analysis
- Trade-off analysis

Risk analysis

In a Business Case perspective, **Risks** are best considered to be threats to the ability of the target to meet the non-functional and security requirements specified as part of the Architecture Definition.

An initial risk analysis should be performed at the beginning of the cycle as well as subsequently as the cycle progresses.

Gap analysis (options)

A **Gap** is a difference between two lists or structures. **Gap Analysis** identifies the difference. A good example of Gap Analysis is comparing Baseline and Target Architectures. This allows identification of elements which need to be removed and added to the Baseline to transform it into the Target state. Further it can highlight when something has been dropped deliberately or by mistake.